



RESEARCH ARTICLE

Enterobacteriaceae community dynamics in sleepy lizards: Richness, prevalence and co-occurrence over time

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Abstract

Animal gut microbiomes can be very diverse, and enteric bacteria can profoundly affect the physiology of their host. The gut microbiome can be related to host health and digestion, which ultimately contribute to host body condition. However, we have a limited understanding of the co-occurrence patterns of gut bacteria in their host, and how co-occurrence and bacterial diversity change over time. This notion is especially important to animals living in groups as bacteria can transmit through social interactions. We investigated the co-occurrence patterns of gut bacteria in a lizard host. We repeatedly collected cloacal swabs from 87 sleepy lizards (*Tiliqua rugosa*) from two different study sites over their activity season. We determined the richness and prevalence of 82 enteric bacterial strains and used a probabilistic model to investigate their co-occurrence. At both study sites, richness and prevalence generally increased over time. We suggest that the lizards acquire strains throughout their activity season by moving through the landscape and inspecting conspecific scats. Lizards continuously tongue-flick while moving, and thereby ingest bacteria when they move through areas where other animals defaecated. Temperature, rainfall and diet change seasonally, influencing lizard activity, and may influence the observed increase in enterobacterial richness and prevalence. Further, albeit with some exceptions, most strain pairs did not occur significantly more often or less often than expected by chance. This finding shows a lack of structured co-occurrence, which may imply that most bacterial strains did not facilitate or inhibit each other. The absence of a co-occurrence pattern could also be driven by random encounters of bacteria shed by other lizards within the habitat. Our results suggest that behaviour (movement patterns, tongue-flicking), activity patterns and environmental factors collectively drive the temporal pattern of the gut bacterial community in sleepy lizards and potentially other wild reptiles.

KEYWORDS

Egerniinae, *Escherichia*, gastrointestinal bacteria, microbial community, *Salmonella*, shingleback lizard

INTRODUCTION

Community dynamics are most commonly studied in macro-organisms; however, micro-organisms also co-occur with each other and establish complex communities, for instance, inside their host (Horner-Devine et al., 2007). In bacterial communities, the focus has been on the co-occurrence of pathogenic and parasitic bacteria, with less attention on commensal bacteria

(Gotelli & Rohde, 2002; Haukisalmi & Henttonen, 1993; Spurgin et al., 2012). Compared to parasites and pathogens, commensal bacteria do not usually cause diseases or draw energy directly from the host (Drew et al., 2021). Nevertheless, they can still impact the host in diverse ways. First, gut commensals can affect the gut–brain axis, thereby influencing the host's physiological processes (Foster & Neufeld, 2013; Sarkar et al., 2016). Second, gut commensals have been shown to influence the central nervous system, and hence can affect host behaviour (Cryan & Dinan, 2012; Davidson et al., 2020). Moreover, mounting evidence suggests that gut commensals play an important role in modulating host immune responses, which can either decrease or increase susceptibility to potentially pathogenic microbes (Brestoff & Artis, 2013; Browne et al., 2017; Fan & Pedersen, 2021). For example, in mice, the gut bacterium *Akkermansia muciniphila* stimulates the immune system and induces a T-cell response (acquired immune response), which helps maintain gut health (Ansaldo et al., 2019). Similarly, other gut commensals promote the maturation of lymphocytes, which may decrease host susceptibility (Chung et al., 2012; Khosravi et al., 2014). However, while commensal bacteria can be involved in important host functions and do not directly harm their hosts, the co-occurrence of gut commensals and potential pathogens can lead to diseases in the host. For instance, gut commensals can promote enteric virus infection by facilitating virus replication, as has been shown in mice (Kuss et al., 2011). Given the different effects co-occurring commensals can have for their host, the first important step is to understand the temporal pattern and potential consistency of commensal co-occurrence and the factors affecting it.

The gut bacterial community within a host can change over time (Schlomann & Parthasarathy, 2019). The environment is one factor that can drive this temporal change (Fontaine et al., 2018; Sepulveda & Moeller, 2020). Different environments have different micro-climates (e.g. shaded vs. open areas). Ambient temperature and sunlight affect the survival time of bacteria within faeces deposited into the environment by their host (Parsons et al., 2015). Furthermore, different bacteria survive at different temperatures. Hence, the environment can modulate the bacteria an individual animal encounters in the habitat. Another factor can be seasonal variation in food availability, which has been shown to influence the host's gut bacterial diversity and composition in wild primates (Gomez et al., 2016; Hicks et al., 2018; Orkin et al., 2019).

A third factor that can structure bacterial communities is interactions among bacterial strains within hosts, including competition, inhibition, facilitation and commensalism (Faust & Raes, 2012). For example, if certain strains are well established in the host already, they may inhibit (or facilitate) the establishment of other strains. A main area of interest is to understand whether co-occurrence patterns are random or structured by these interactions (Vellend et al., 2014). Because it is challenging to observe interactions within communities in the natural environment directly, the outcome of these interactions is commonly used to make inferences about those interactions (Björk et al., 2018; Diamond, 1975; Gotelli et al., 2010; Stone & Roberts, 1990). For example, if two bacterial strains co-occur more frequently than expected by chance (a positive association), one possible explanation could be facilitation and/or mutualism, while a negative association (if they co-occur less often than expected by chance) could result from competition and inhibition (Faust & Raes, 2012). On the other hand, co-occurrence patterns within a community can also be explained by species specific responses to the environment (Peres-Neto et al., 2001), and their habitat preference. Shared preferences can then lead to positive associations, whereas different preferences result in negative associations among individual dyads (Estevo et al., 2017). For instance, the physiological condition (e.g. temperature, pH) of some hosts may be favoured by certain bacteria,

and hence these bacteria are more likely to co-occur in these hosts (Levy & Borenstein, 2014). These inferences predominantly rely on the assumption that the co-occurrence between individuals is non-random in the community (Gotelli, 2000). However, stochastic processes can also play an essential role in shaping co-occurrence patterns (Chave, 2004). For instance, variation in microbe replication and accidental deaths (Ofițeru et al., 2010) has been shown to affect the microbial community structure and promote the coexistence of different taxa (Nemergut et al., 2013; Zhou & Ning, 2017). A stochastic process can be inferred if the co-occurrence pattern of most bacterial dyads does not differ from what is expected by chance (Williams et al., 2014). Moreover, a consistent co-occurrence pattern over time may provide evidence of an underlying community structure (Williams et al., 2014).

Bacteria from the family Enterobacteriaceae have been recorded in various vertebrates worldwide, including many reptilian species in Australia (Gordon & Cowling, 2003; Parsons et al., 2011). This family contains bacteria responsible for bacterial infections transmitted from reptiles to humans, such as *Salmonella* spp. serovars (Ebani, 2017; Kikillus et al., 2011; Shnaiderman-Torban et al., 2019). Nevertheless, their temporal variation (Blyton et al., 2013) and co-occurrence pattern within their wild host are rarely studied. The Australian sleepy lizard (*Tiliqua rugosa*) hosts *Escherichia coli* and various strains of *Salmonella enterica* (Bull et al., 2012; Gordon & Cowling, 2003), which are considered commensal in this lizard. The transmission of *S. enterica* subspecies was correlated with microhabitat factors, and home range overlap among lizards (Bull et al., 2012; Parsons et al., 2015). Thus, bacterial diversity is likely to be subject to the seasonal activity of sleepy lizards. The main activity period of the sleepy lizard extends from the Austral early spring to mid-summer. In winter, during aestivation, the lizards generally do not eat, so the prevalence of enteric bacteria such as *Salmonella enterica* is typically low in early spring when lizards become active (Parsons et al., 2015). Sleepy lizards are omnivorous and feed on herbaceous plants, including annual flowers and occasionally invertebrates (Dubas & Bull, 1991). Food resources become scarce in summer. This change in food availability may also result in fluctuations in gut bacteria diversity. Therefore, sleepy lizards are ideal for studying the temporal variation in Enterobacteriaceae bacteria and also how different strains co-occur within the host.

Here, we investigated the temporal dynamics and co-occurrence of commensal Enterobacteriaceae strains in sleepy lizards. We first quantified bacterial diversity and prevalence in our study population and whether they changed over time (i.e. whether more lizards carried more diverse bacterial strains at different times of the activity season). We also tested whether bacterial diversity and prevalence was related to body condition (proxy for health) or sex. Then, we investigated the co-occurrence patterns of bacterial strains to determine whether certain strains co-occur non-randomly. Finally, we determined whether the co-occurrence patterns of bacterial strains were consistent over time. Our study will contribute to a better understanding of the role enteric bacteria play in lizards and whether deterministic processes such as facilitation and inhibition shape enteric bacterial composition.

METHODS

Study system and sampling

We conducted fieldwork near Bunday Bore Station, Australia (33°40'S, 139°20'E), from spring (late August) to summer (early December) 2014, which is within the activity season of the sleepy lizard. We tracked 87

lizards from two separate sites (site A: approx. 3.09 km²; site B: approx. 2.55 km²). Twenty-two males and 21 females were caught from site A, and 27 males and 17 females were caught from site B, respectively. Lizards from the two sites had been part of a long-term research project on parasite transmission and behavioural ecology. Lizards did not travel between the two sites (unpublished GPS tracking data) but belonged to the same meta-population. The distance between the two sites was approximately 2.2 km, which is further than the span of a typical home range (0.04 km²) (Bull & Freake, 1999). Vegetation at both sites was dominated by chenopod shrubs and pearl bluebush (*Maireana sedifolia*). Patches of *Casuarina cristata* woodland provided shade for the lizards in the hot, dry summer. Most *Casuariana* patches were at site A, which was more shaded than site B.

We attached a unit consisting of a radio transmitter and a GPS device to each lizard's tail using surgical tape. The unit weighed 37 g, which was 5.3% of an average 693 g lizard and 7.7% of the lightest 480 g lizard in this study. Previous studies have used similar units during many earlier projects and did not detect any effects on lizard behaviour or skin (Bull et al., 2012; Leu et al., 2010, 2011; Payne et al., 2020; Spiegel et al., 2015, 2017, 2018; Taggart et al., 2018).

During the study, each unit's unique radio transmitter frequency allowed us to locate and sample each lizard every fortnight. We captured lizards by hand and the handling time was usually 15 min (maximum approximately 30 min). Sampling the majority of lizards at each site typically took 3 days. We sampled the two sites immediately after each other and kept the sampling order consistent. We swabbed the cloaca of each lizard using a sterile swab/transport tube system containing Amies transport agar (Copan M40). Although cloacal samples are more likely to represent the hind gut bacterial diversity, they may encompass the breadth of bacterial diversity in a reptile's gut (Colston et al., 2015; Kohl et al., 2017). The swabs were stored at 4°C before being shipped to the Australian National University for processing and bacterial strain identification (approximately 4 days on average from sampling to processing). At the end of the study, we removed the units from all lizards. Some lizards could not be caught every fortnight, for example, if they were in a wombat burrow. On average, 90% of the lizards were caught and swabbed (78/87) each fortnight (sampling point), resulting in a total of 468 swabs. Every lizard was sampled at least four times, and 49% were sampled on all six sampling points.

Bacterial species and strain identification

Bacteria were grown from the samples by inoculating each swab into 10 mL of MacConkey broth (incubated at 35°C for 18 hours with shaking) (Power & McCuen, 1988). The cells from 1 mL of each culture were then pelleted by centrifugation, resuspended in 1.5 mL of lysogeny broth and glycerol (2:1), and stored at −80°C. DNA was then extracted from 100 µL of the cell freezer cultures using the Isolate II genomic DNA kit (Bioline) with 1-hour digestion, according to the manufacturer's instructions. All samples were cultured and processed in aerobic conditions.

To identify the species and strains carried by each lizard at each time point, a 502 bp region of the RNA polymerase beta-subunit encoding gene (*rpoB*) was amplified from the DNA extracts and sequenced. The *rpoB* gene was chosen for strain identification as it is a core housekeeping gene and has previously been shown to give greater taxonomic resolution within the family Enterobacteriaceae than the 16S rRNA gene (Mollet et al., 1997).

Prior to metataxonomic sequencing of the samples, we checked that all Enterobacteriaceae species present in the samples could be amplified

using our primer set by isolating 142 bacterial clones from 82 different lizards. To isolate the different enteric bacterial clones from the mixed culture samples, we streak plate method each sample onto a MacConkey agar plate, which was then incubated at 37°C overnight. One representative colony of each unique colony morphology observed on a plate was then dilution streaked onto another MacConkey agar plate (then incubated at 37°C overnight). DNA was then extracted from these unique bacterial clones by inoculating a single colony from the second MacConkey agar plate into 5 mL of lysogeny broth, which was incubated for 18 hours at 35°C with shaking. DNA was then extracted using the Isolate II genomic DNA kit (Bioline) as above. Full genome sequencing of the isolates was then performed following Blyton and Gordon (2017) and the species identified using the KmerFinder tool provided by the Centre for Genomic Epidemiology (<http://www.genomicepidemiology.org/>) (Hasman et al., 2014; Larsen et al., 2014) (Table S1). This analysis confirmed that all Enterobacteriaceae genera and species except for the basal *Proteus* could be amplified using our primers. Only a single *Proteus* was identified among the clones and was therefore considered a minor component of the lizards' enteric community (Table S1). We excluded *Proteus* from the analyses.

The *rpoB* gene was amplified from the mixed culture DNA extracts following Illumina's '16S Metagenomic Sequencing Library Preparation' protocol (Part # 15044223 Rev. B) using custom primers (RpoB2.F: GGACCARAACAACCC; RpoB2.R: ACCCATCAATGCACG) (Amplicon et al., 2013). We designed the primers by aligning the core genomes of the 142 bacterial clones and identifying areas of the genomes with maximum variation (low identity among isolates) to give high resolution. The primers were not specific to Enterobacteriaceae, and hence may amplify other species of bacteria. Hence, we used selective media to grow the bacteria to minimize the detection of non-target families. Sequencing libraries were prepared from the PCR products using the Nextera XT sample preparation kit (Illumina) following the manufacturer's protocol. We generated a library for each sample. Each sample was uniquely tagged using the Nextera XT indexing primers (Illumina), and 96 libraries were pooled in a sequencing run. Paired-end sequencing was performed at the Biomolecular Resource Facility, Australian National University, on the Illumina Miseq using the version 3 reagent kit for 300 cycles (read length: 2 × 300 bp). A PCR negative control was included on each plate to verify low contamination rates.

We removed barcode and primer sequences from the returned sequencing reads in CLC genomic workbench 7.0 (QIAGEN). The reads were then trimmed for quality and paired-reads merged in CLC using default parameters. Reads that were not the correct length were then removed. The processed reads were then clustered at 99% nucleotide similarity into unique operational taxonomic units (OTUs) using QIIME2 (Caporaso et al., 2010). The resulting OTU by sample table was rarefied to 10 000 reads per sample using the R package vegan (Oksanen et al., 2018). Any OTUs found in only a single lizard at a single time point were then discarded. The species identity of each unique *rpoB* OTU was determined using Microbial Blast at NCBI. These OTUs were identified as different bacterial strains in this study.

The relative abundance of the identified strains, as estimated from the number of sequencing reads, was not used in the analysis of the sleepy lizards' microbial communities as the strains' abundance in the mixed cultures may not have reflected their abundance in the lizards due to the culturing step. Nonetheless, sequence analysis of three mock communities that were assembled by combining pure cultures in known ratios showed that strains that were present in mixed cultures at a relative abundance of 0.01% were detected in over 90% of cases (Table S2). The mock

communities were assembled by growing individual isolates from different species separately in lysogeny broth and then pooling them in known ratios based on cell density.

Bacterial diversity and prevalence

To understand variation in bacterial diversity, we transformed the reads into a presence/absence table. The presence or absence of each bacterial strain was recorded for each lizard per fortnight.

First, based on these data, we calculated bacterial richness for every sampling point as the number of bacterial strains that a lizard carried. We investigated whether the richness of bacterial strains within each lizard changed over time using a generalized linear mixed model with Poisson distribution. *Richness* was the dependent variable, while *sex*, *site*, were fixed factors, and *body condition*, *sampling point* were continuous variables. Lizard ID was added as a random factor to account for the repeated measures design of the study. We added sex into the model because males and females often differ in their susceptibility to becoming infected due to the potential immuno-suppressive effect of testosterone (Zuk, 2009). This study contained lizards from two separate study sites, and we included the *site* as a fixed factor. Body condition can be related to gut bacterial diversity, and hence we included it as a continuous variable (Davidson et al., 2021; Mshelia et al., 2018; Ruiz-Rodríguez et al., 2009). We calculated the body condition as the residual of a mixed-model regression of log-transformed SVL against log-transformed body mass, with a random factor of sampling points (Godfrey et al., 2013). We used the sampling point as a continuous variable in the model to investigate the overall change of bacterial richness over time. Age was not included in the model as the age of the sleepy lizards was not known, though all lizards were adults.

Second, we calculated bacterial prevalence as the proportion of lizards that carried a particular strain per sampling point. Bacterial prevalence was calculated separately for each sex and each site. We analysed the prevalence of each bacterial strain over time using a zero-inflated generalized linear mixed model with binomial distribution. *Prevalence* was the dependent variable, and *sex*, *site*, were fixed factors, and *sampling points* was a continuous variable. *Bacterial strain number (OTU ID)* was added as a random factor to account for the repeated sampling design.

Co-occurrence of bacterial strains

For each sampling point separately, we determined the co-occurrence of all possible pairs (dyadic combinations) of bacteria strains using a probabilistic model (Veech, 2013; Veech & Araújo, 2014). We used the 87 hosts as individual sites in which two bacterial strains might both occur. This model calculates an expected frequency of co-occurrence between each pair of strains based on chance alone but accounting for the prevalence of bacterial strains among the hosts. This approach thereby corrects for the expected co-occurrence under the null model using the Jaccard's index. This is particularly useful when the prevalence of bacterial strains is high (>75%) or low (<10%) (Fountain-Jones et al., 2020; Mainali et al., 2017). The probability model compares the observed frequencies of co-occurrence among bacterial strains with the respective expected frequencies. The algorithm returns a positive association between a pair of strains if the observed frequency is significantly greater than expected by chance and a negative association if the observed frequency is significantly lower than

expected by chance. We conducted the co-occurrence analysis using the R package *cooccur* (Griffith et al., 2016). We then used a chi-squared test to determine whether positive associations occurred more often than negative associations among the six sampling points.

Bacterial co-occurrence pattern over time

To understand the consistency of strain dyads over time, we examined the relationship of co-occurrence matrices between sampling points. First, we calculated the effect size as the difference between the expected and observed frequency of co-occurrence for a given dyad of strains (Veech, 2013). Then, following Griffith et al. (2016), we divided the difference by the number of lizards sampled at each sampling point as the standardized effect size. For each sampling point ($N=6$), we generated a strain-by-strain matrix where each cell represented this measure. We used multiple regression quadratic assignment procedures (MRQAP) to test whether the co-occurrence matrices were consistent through time. The MRQAP is a multiple regression analysis for matrices, which tests the relationship between the dependent matrix and each independent matrix while controlling for all other independent matrices. In our study, the matrix of the last sampling point was the dependent matrix, and the matrices of the earlier sampling points were the independent matrices. For example, $\text{sampling point } 3 = \beta_1 \times \text{sampling point } 1 + \beta_2 \times \text{sampling point } 2$. We used the function *mrqap.dsp* in the package *asnipe* in R (Farine, 2013). The regression coefficient for each sampling point was calculated using the Double-Semi-Partialing (DSP) method (Dekker et al., 2007). It minimizes the correlation between the focal independent matrix and the other independent matrices. The MRQAP includes a permutation procedure that randomizes the regression residuals of the independent matrix, which is used to calculate the p-value. For this procedure, we conducted 1000 permutations. We then built co-occurrence networks based on the positive and negative associations among bacterial strains using the bacterial strains as the node and the effect size as the edge weight. To understand if certain bacteria can have a strong influence over other strains, we identified the most influential strains using the eigenvector centrality, which measures the number of high degree centrality nodes a node is connected with.

RESULTS

Bacterial strains identification

We took a total of 468 cloacal samples (swabs) from the 87 sleepy lizards across the six sampling points. Amplicon sequencing of the *rpoB2* gene amplified from DNA extracted from culturing the swab samples returned between 218 and 166583 reads per sample (mean=32143 reads per sample). A total of 429 samples were retained after samples with fewer than 10000 reads were removed from the dataset. All remaining samples were rarefied to 10000 reads. We identified 82 different bacterial strains (OTUs) in the cloaca of our sleepy lizard study populations (Table S3). The most common strains belonged to the genera *Salmonella*, *Citrobacter* and *Escherichia*, which were detected in 100%, 99.5% and 95.3% of all samples, respectively. Although our study mainly aimed to investigate the co-occurrence of Enterobacteriaceae, we also detected some bacteria outside this family (e.g. Hafniaceae, Morganellaceae and Yersiniaceae).

and even a few not belonging to the order Enterobacterales, that is, the orders Aeromonadales and Pseudomonadales (Table S3). We included all strains detected in our analyses and we stress that ‘bacterial richness’ in this study was restricted to bacterial strains mentioned in Table S3.

Bacterial diversity and prevalence

Both richness and the overall prevalence significantly increased over the six sampling points (Table 1A: $\beta=0.06$, $p<0.001$; Table 1B: $\beta=0.12$, $p=0.04$; Figure 1). We found no relationship between body condition and bacterial richness of sleepy lizards ($\beta=0.11$, $p=0.50$). Neither bacterial richness nor prevalence differed between study sites A and B (Table 1A: $\beta=0.06$, $p=0.09$; Table 1B: $\beta=0.11$, $p=0.57$), or between sexes (Table 1A: $\beta=-0.05$, $p=0.13$; Table 1B: $\beta=-0.08$, $p=0.67$).

Co-occurrence of bacterial strains

We analysed a total of 3321 strain dyads per sampling point in our dataset. The number of positively associated strain dyads ranged from 9 (0.27%) to 48 (1.45%), and the number of negatively associated ones ranged from 2 (0.06%) to 10 (0.30%). Positive associations did not occur significantly more often than negative associations at each sampling point ($\chi^2=2.16$, $df=5$, $p=0.83$, Table 2). However, the associations among strains were predominantly random (up to 99.7%), and most dyads did not co-occur more often or less often than expected if the strains were distributed randomly in the lizard populations (Figure 2). We also found the associations of the most influential strains (i.e. highest eigenvector centrality) were generally among strains from different genera. The most influential strain varied among the sampling points (Figure S1). These strains were *Salmonella* strain 3 at sampling point 1, *Salmonella* strain 3 and *Enterobacter* strain 2 at sampling point 2, *Citrobacter* strain 4 at sampling point 3, *Cronobacter* strain 4 at sampling point 4, *Klebsiella* strain 4 and *Serratia* strain 2 at sampling point 5, *Enterobacter* strain 1 and *Hafnia* strain 1 at sampling point 6.

TABLE 1 The effect of site and sampling point on richness (A) and prevalence (B).

A. Richness				
Fixed effects	Estimate	95% Confidence interval	Wald χ^2	<i>p</i> -value
Sampling point	0.06	0.04–0.08	36.67	<0.001
Body condition	0.11	–0.20–0.42	0.46	0.50
Sex (Female)	–0.05	–0.12–0.02	2.28	0.13
Site (B)	0.06	–0.01–0.12	2.81	0.09
B. Prevalence				
Fixed effects	Estimate	95% Confidence interval	Z-value	<i>p</i> -value
Sampling point	0.12	0.003–0.229	2.02	0.04
Sex (Female)	–0.08	–0.46–0.30	–0.42	0.67
Site (B)	0.11	–0.27–0.49	0.56	0.57

Note: Richness and prevalence increased significantly over the sampling points. *p*-values under 0.05 were marked in bold.

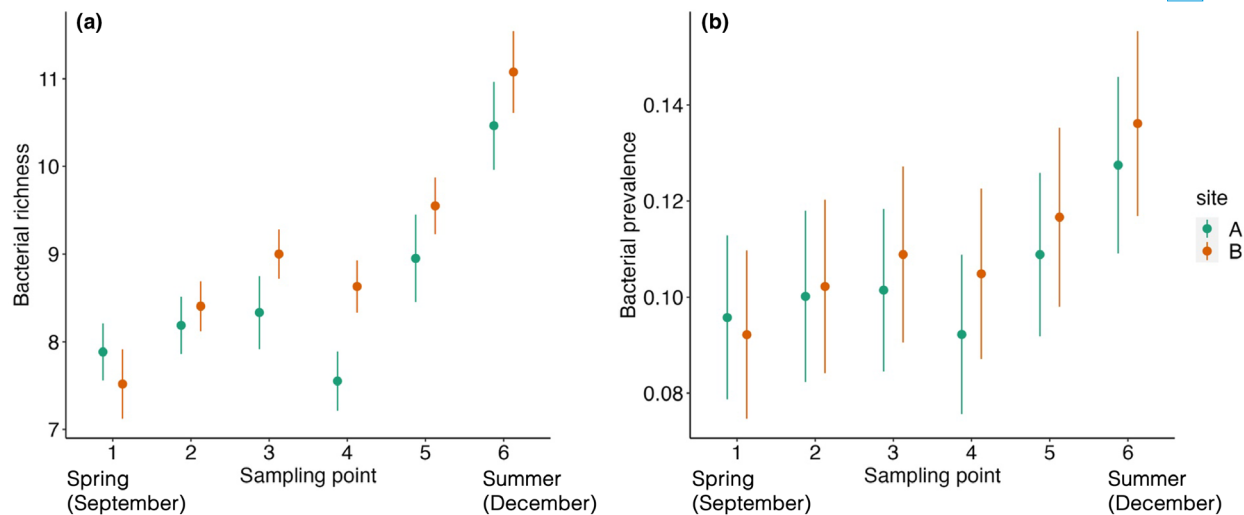


FIGURE 1 Point-range plots of bacterial richness (the number of strains that a sleepy lizard carried) (a), and bacterial prevalence (the ratio of lizards that carry a certain bacterial strain) (b) over time (sampling point) and between sites. Sampling points occurred approximately every fortnight. The solid circle represented the mean of bacterial richness at each sampling point, and the error bars represented one standard error.

TABLE 2 The number of positive, negative and random bacterial associations for each sampling point.

Sampling point	Positive associations	Negative associations	Random
	Number (%) of dyads	Number (%) of dyads	Number (%) of dyads
1	9 (0.27)	2 (0.06)	3310 (99.67)
2	22 (0.66)	2 (0.06)	3297 (99.24)
3	26 (0.78)	6 (0.18)	3289 (99.04)
4	14 (0.42)	4 (0.12)	3303 (99.46)
5	48 (1.45)	10 (0.30)	3263 (98.25)
6	40 (1.20)	6 (0.18)	3275 (98.61)

Bacterial co-occurrence temporal pattern

Our MRQAP analyses showed that the six matrices of co-occurrence were significantly related over time. The co-occurrence measure at the previous sampling point best explained the co-occurrence among strain dyads at the current sampling point (Table 3). However, the overall correlations between matrices were relatively weak (Adjusted R^2 range: 0.0005–0.1158, Table 3). We also did not find any co-occurrence pattern among the most influential strains (Appendix I; Figure S1A–F).

DISCUSSION

Investigating bacterial diversity and whether bacteria affect each other's presence/absence is important for understanding host–bacteria relationships. We investigated this in the sleepy lizard and focused on the bacterial family Enterobacteriaceae, plus some additional strains of other families. We deduced associations among bacteria by examining the variation in bacterial diversity and their co-occurrence pattern over time

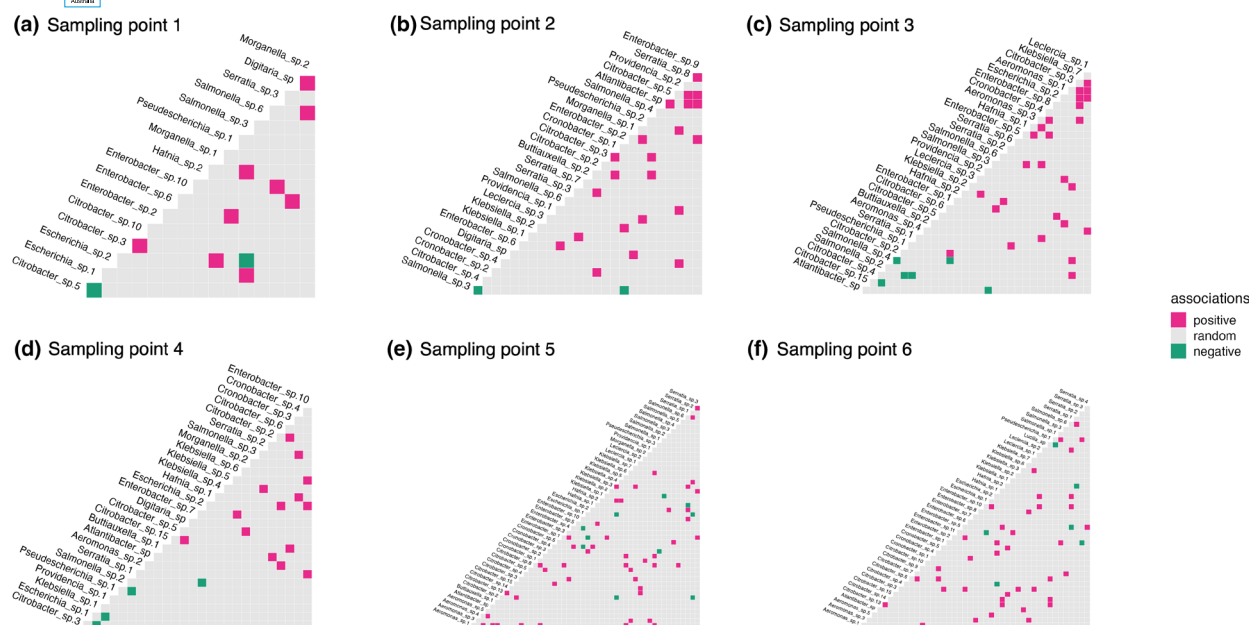


FIGURE 2 Matrices showing the co-occurrence pattern. Most dyads co-occurred randomly in sleepy lizards. The x-axis and y-axis represent each operational taxonomic units whose taxon is marked above the axis. The significant strain dyads were converted into positive or negative associations according to their effect size. Positive associations are coloured in purple, negative associations are coloured in green. Grey colour represents strain dyads whose co-occurrence frequency does not differ from that estimated by chance. (a) Sampling point 1. (b) Sampling point 2. (c) Sampling point 3. (d) Sampling point 4. (e) Sampling point 5. (f) Sampling point 6.

TABLE 3 The estimates (upper triangular) and p values (in brackets) calculated by multiple regression quadratic assignment procedures.

Sampling point	Estimates (p_{rand})					
	1	2	3	4	5	6
1		0.025 (0.107)	0.0005 (0.965)	−0.056 (<0.001)	0.016 (0.370)	0.028 (0.174)
2	—		0.113 (<0.001)	0.034 (0.063)	0.034 (0.094)	−0.071 (0.009)
3	—	—		0.106 (<0.001)	0.094 (<0.001)	0.028 (0.187)
4	—	—	—		0.160 (<0.001)	0.133 (<0.001)
5	—	—	—	—		0.362 (<0.001)
6	—	—	—	—	—	
Intercept		2.72×10^{-6}	1.68×10^{-4}	-9.48×10^{-6}	4.89×10^{-4}	4.91×10^{-4}
Multiple R^2		0.0008	0.0057	0.0178	0.0290	0.1172
Adjusted R^2		0.0005	0.0099	0.0169	0.0278	0.1158

Note: The co-occurrence matrix of each sampling point was regressed against its matrices of preceding sampling points. The intercept, multiple r -square and adjusted r -square of each corresponding model were shown below the matrix. Significant sampling points were marked in bold.

from September to December. Our study has two major findings. First, bacterial richness within individual sleepy lizards and prevalence within the two populations increased over time. Second, we identified only a few positive and negative co-occurrence associations among strain dyads. The co-occurrence of the majority of strains was not different from what was expected by chance.

Variation in bacterial richness and prevalence

The increase in bacterial richness and prevalence over time is consistent with findings from Parsons et al. (2015) who reported that subspecies

of *Salmonella enterica* increased in their abundance over the course of the sleepy lizard activity season. Sleepy lizards aestivate inside burrows during the summer heat and typically remain inactive during autumn and winter (Bull, 1987; Kerr et al., 2003; Satrawaha & Bull, 1981). Hence, they rarely feed between January and August, the period before our study. Aestivation or hibernation can reduce bacterial diversity in winter, which has been shown in ground squirrels (Dill-McFarland et al., 2014), brown bears (Sommer et al., 2016) and bats (Xiao et al., 2019). The low diversity measures at the start of our study, which was the start of the activity season of the lizard, suggest that this is also the case in sleepy lizards. Then, gut bacteria accumulate again as lizards are active. In particular, lizard activity and social interactions facilitate the transmission of gut bacteria (Bull et al., 2012), which could explain this pattern of accumulation over time. When moving through the environment, individuals may encounter contaminated areas such as latrines (Fenner et al., 2015). Lizards continuously tongue-flick the ground to inspect their environment using chemical cues. Individuals that tongue-flick in these contaminated areas could ingest bacteria. Although there could be some cold-adapted bacteria (e.g. *Yersinia*, *Campylobacter*, *Arcobacter*) that may overwinter with the lizards (Gilbert et al., 2019; Palonen et al., 2010), our bacteria culture methods were selective to Enterobacteriaceae and we did not find any of our studied bacterial strains become less likely to be present or less prevalent over the activity season.

An alternative explanation of the increased bacterial diversity is that diet differences across the season (which we did not measure) may also have influenced bacterial richness and prevalence. Sleepy lizards change their diet from flowers and berries to succulent leaves during summer (November – January) (Dubas & Bull, 1991). In summer, food resources may become patchier than in spring. Whether sharing these resources influences gut bacteria presence and diversity is unknown. Similar to water resources, contaminated food patches may also increase the possibility of acquiring bacterial strains. An earlier study found that lizards that shared space also shared the same bacterial strains (Bull et al., 2012). This finding may suggest that resource sharing may play a role in bacterial transmission in the lizards. Similarly, overnight refuge sharing explains the infestation with ectoparasitic ticks in the sleepy lizard (Leu et al., 2010).

We showed that male and female lizards did not differ in bacterial richness or prevalence. Similarly, three other lizard species (*Liolaemus parvus*, *L. ruibali* and *Phymaturus williamsi*) from the Andes Mountains also showed no sex difference in bacterial diversity (Kohl et al., 2017). However, other studies have found sex differences in gut bacterial diversity, for instance, in the striped plateau lizard (*Sceloporus virgatus*) (Martin et al., 2010). These findings suggest that sexes may not always differ in bacterial diversity. In sleepy lizards, males and females share the inner 50% core home range and refuge sites (Kerr & Bull, 2006; Leu et al., 2015). These spaces use similarities and strong pair behaviour could have homogenized the gut bacteria composition between males and females in our sleepy lizard population.

Co-occurrences of bacterial strains within lizards

The majority of bacterial strain dyads were neither more nor less frequently detected than expected by chance, that is, they did not differ from random. If two strains co-occurred more frequently than estimated by chance, we interpreted these associations as facilitation. Conversely, if two strains co-occurred less frequently than estimated, we interpreted this association as

inhibition or avoidance. Nevertheless, positive and negative associations were very rare, suggesting a lack of structured co-occurrences.

We showed that the co-occurrence pattern between bacterial strain dyads was correlated over time. For example, the co-occurrence pattern of sampling point 6 was best explained by the pattern of sampling point 5. Earlier sampling points were progressively less predictive. This result could suggest that there was a small but continuous change from sampling point (fortnight) to sampling point. However, the magnitude of change differed among dyads. This notion is further supported by a low *r*-square value across all sampling points. It is also in agreement with our finding that richness and prevalence increased over time. The lack of non-random co-occurrence may lead to the persistence of diverse bacterial strains in the host, which increased over time during the activity season. Our study focused on the co-occurrence of mainly Enterobacteriaceae but the diversity can also be affected by bacteria from other families or from other orders. Although some families other than Enterobacteriaceae were included in this study, we did not find any evidence that these bacteria were associated with Enterobacteriaceae diversity.

Ecological interaction among taxa is known to be a strong force influencing microbial assemblages. Other studies have found that facilitative (He et al., 2014; Li et al., 2019) and competitive interactions (Friedman et al., 2017) can shape community structure. This was not the case in our study, where most strain dyads co-occurred randomly. Co-occurrence patterns can also emerge through shared characteristics and habitat preferences. For example, positive associations could result if different strains persist in the same habitat, and show similar responses to environmental factors, such as heat inducing desiccation, instead of biotic interactions (Blanchet et al., 2020; Cazelles et al., 2016; Dormann et al., 2018; Hirano & Takemoto, 2019). External processes and factors, such as the inspection of contaminated areas through tongue-flicking, coupled with divergent off-host survival times among strains, and environmental conditions, including the amount of shade within the host's home range, may influence the co-occurrence of gut bacteria. The scatting behaviour of sleepy lizards, which defecate haphazardly in different microhabitats inside their home range (Fenner et al., 2015), could influence the off-host survival times depending on scat placement (Parsons et al., 2015). Thus, this behaviour combined with external factors may facilitate sleepy lizards ingesting bacterial strains from the environment when they visit and inspect faeces of other individuals in different microhabitats. Furthermore, human-altered environments may change gut microbial diversity through geographic isolation or exposure to domestic animals, which has been demonstrated in birds and mammals (Grange et al., 2016; Heni et al., 2023). Lizards in our study sites shared the habitat extensively with managed sheep. Sheep densities were low and consistent throughout our data collection period. Whether lizards can obtain bacteria from livestock remains unstudied. There are some examples that diverse gut bacteria can benefit the host by facilitating immune responses, energy intake and general vigour (Chen et al., 2019; Maltby et al., 2013; Ray et al., 2012). These physiological responses can have far-reaching consequences for host performance, such as movement pattern and cognitive performance (Davidson et al., 2020; Schretter et al., 2018). However, we did not find any association with gut bacteria diversity and body condition in our study.

Our study showed that the Enterobacteriaceae community changed over time. This result highlights the importance of the ecological environment in this host–bacteria relationship. The environment could either directly impact the bacteria, for instance through differential survival time when off host, or indirectly through different host behaviour at different times of its

activity season (Somers et al., 2023). A combination of both is also conceivable. The absence of a clear co-occurrence structure may further suggest the influence of external factors in maintaining bacterial diversity in this system. However, during periods of inactivity, lizards are likely to lose the bacteria inside their digestive system. Hence, further investigations of host interactions with the ecological environment are needed to better understand bacterial diversity in hosts.

AUTHOR CONTRIBUTIONS

Ko-Huan Lee: Conceptualization (equal); data curation (equal); formal analysis (equal); methodology (equal); validation (equal); visualization (equal); writing – original draft (equal); writing – review and editing (equal). **Michaela D. J. Blyton:** Data curation (equal); formal analysis (equal); investigation (equal); methodology (equal); validation (equal); writing – review and editing (equal). **Stephanie S. Godfrey:** Conceptualization (equal); funding acquisition (equal); writing – review and editing (equal). **Andrew Sih:** Conceptualization (equal); funding acquisition (equal); writing – review and editing (equal). **Michael G. Gardner:** Conceptualization (equal); funding acquisition (equal); writing – review and editing (equal). **Martin J. Whiting:** Resources (equal); supervision (equal); validation (equal); writing – review and editing (equal). **Stephan T. Leu:** Conceptualization (equal); funding acquisition (equal); investigation (equal); methodology (equal); project administration (equal); supervision (equal); validation (equal); writing – review and editing (equal).

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data and supplementary materials of this study are available in the Open Science Framework (OSF) at https://osf.io/a7tw3/?view_only=44c54ee833984026973bea0208a37980.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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